QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/402A
9 AUGUST 1983
SUPERSEDING
MIL-M-38510/402
14 June 1982

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, MOS,
1024-BIT STATIC RANDOM ACCESS MEMORY (RAM),
MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, N-channel static MOS, 1024-bit RAM microcircuits. One product assurance class and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part complete number shall be as specified in MIL-M-38510, with the exception that the "JAN" or "J" certification mark shall not be used.

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit organization package	Address access time
01 (T_C = -55°C instant-on to +125°C) 1/	128 words/8-bit	TAVQV = 450 ns

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline see MIL-M-38510, appendix C
J	D-3 (24-lead, 1/2" x 1-1/4") dual-in-line package

1.3 Absolute maximum ratings.

Input voltage range, all inputs	-0.3 V to +7.0 V
Storage temperature range	-65°C to +150°C
Case operating temperature range (T_C)	-55°C to +125°C
Supply voltage range	-0.3 V to +7.0 V 2/
Power dissipation (P_D)	1.0 W maximum
Lead temperature (soldering, 5 seconds)	270°C
Thermal resistance, junction-to-case (θ_{JC})	82.5°C/W
Maximum junction temperature (T_J)	150°C

1/ $T_C = T_A$ at test time equals zero. Instant-on is defined as all functional characteristics guaranteed at all temperatures 50 ms after power is applied.

2/ Under absolute maximum ratings, the voltage values are with respect to the most negative supply voltage, V_{SS}. Throughout the remainder of this specification, the voltage values are with respect to V_{SS}.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5962

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1.4 Recommended operating conditions.

	Min	Max	Unit
Supply voltage range	4.5	5.5	V
Low level input voltage range (V_{IL})	-0.3	0.8	V
High level input voltage range (V_{IH})	2.0	V_{CC}	V
Read or write cycle time (TAVAV)	450	---	ns
Address setup time (TAVSX)	20	---	ns
Address hold time (TSXAX)	0	---	ns
Read to chip select time (TWHSX)	0	---	ns
Read hold time from chip select (TSXWX)	0	---	ns
Chip select pulse width (TSVSX)	300	---	ns
Write to chip select time (TWLSX)	0	---	ns
Data setup time (TDVSX)	190	---	ns
Data hold time from chip select (TSXDX)	10	---	ns
Write hold time from chip select (TSXWH)	0	---	ns

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Block diagram. The block diagram shall be as specified on figure 2.

3.2.3 Truth table. The truth table shall be as specified on figure 3.

3.2.4 Functional tests. The functional tests used to test this device are contained in the appendix. If the test patterns cannot be implemented due to test equipment limitations, alternate test patterns to accomplish the same results shall be submitted to the contracting activity for approval.

3.2.5 Case outline. The case outline shall be as specified in 1.2.2 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \leq T_C \leq +125°C$	Limits Min	Limits Max	Unit
Low level input leakage current (all input pins)	I_{IL}	V_{CC} = 5.5 V, V_{IN} = GND		2.5	µA
High level input leakage current (all input pins)	I_{IH}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		2.5	µA
Output leakage current (tri-state)	I_{LO}	$\overline{CS}$ = V_{IL}, V_{CC} = 5.5 V, $\overline{CS}$ = V_{IH}, D_O = 0.4 V and 2.4 V, all other pins grounded		10	µA
Power supply current	I_{CC}	V_{CC} = 5.5 V, all other pins grounded		100	mA
Low level output voltage	V_{OL}	I_{OL} = 1.6 mA		0.4	V
High level output voltage	V_{OH}	I_{OH} = -205 µA	2.4		V
Output capacitance 2/	C_o	V_o = 0 V, f = 1 MHz, T_C = 25°C, CS_ϕ = 0		12.5	pF
Input capacitance 2/	C_i	V_{IN} = 0 V, T_C = 25°C, f = 1 MHz		7.5	pF
Address access time	TAVQV	See table III (dynamic load is per figure 5)		450	ns
Chip select access time	TSVQV			230	ns
Data hold from end of read	TWXQZ		10	80	ns
Data valid from end of CS	TSXQZ		10		ns
Data hold from end of address change	TAXQZ		10		ns

1/ Output levels are tested in static state and are specified over voltage range of V_{CC}.

2/ This parameter is not tested in table III.

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3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container. The "JAN" or "J" certification mark shall not be used.

3.7 Manufacturer eligiability. To be eligible to supply microcircuits to this specification, a manufacturer shall have a manufacturer certification in accordance with MIL-M-38510 for at least one line. (Not necessarily the line producing the device type described herein.)

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) Class B devices
Interim electrical test parameters (pre burn-in) (method 5004)	2, 8*
Final electrical test parameters (method 5004)	1**, 2, 3, 7**, 8
Group A test requirements (method 5005)	1, 2, 3, 7, 8, 9, 10, 11
Group C end-point electrical parameters (method 5005)	NA
Group D end-point electrical parameters (method 5005)	1, 2, 3, 7, 8,
Additional electrical subgroups for group C periodic inspection	NA

* Maximum temperature only.

**PDA applies to subgroups 1 and 7 (see 4.2d).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspections. The following additional criteria shall apply:

a. When the alternate screening option of 3.3 of method 5004 is applied to class B devices under this specification, the following additional items are applicable:

(1) Internal visual, method 2010, condition B. In addition to the changes indicated by 3.3.1 of method 5004, the following additional clarifications and deletions are applicable:

(a) Top layer conductors or top poly-silicon conductors shall be inspected using the metallization criteria of MIL-STD-883, method 2010. Underlying conductors such as poly-silicon need not be inspected.

(2) A cell bit stress test may be used for the special electrical screen test of 3.3.1 of method 5004. The cell bit stress test plan shall be submitted to and approved by the preparing activity. The functional test shall be checkerboard/checkerboard algorithm of the appendix. This test shall be performed once at the first high temperature (125°C) functional test only.

b. Burn-in test (method 1015 of MIL-STD-883, test condition D or F) using the circuit shown on figure 4, or equivalent; T_A = 125°C minimum.

c. Interim and final electrical test parameters shall be as specified in table II herein, except the interim electrical test parameters prior to burn-in are optional at the discretion of the manufacturer.

d. The percent defective allowable for class B devices shall be 10 percent based on failures from group A, subgroup 1 test, after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in, divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.6) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6 and 8 of table 1 of method 5005 of MIL-STD-883 shall be omitted.

Text continues on page 16.

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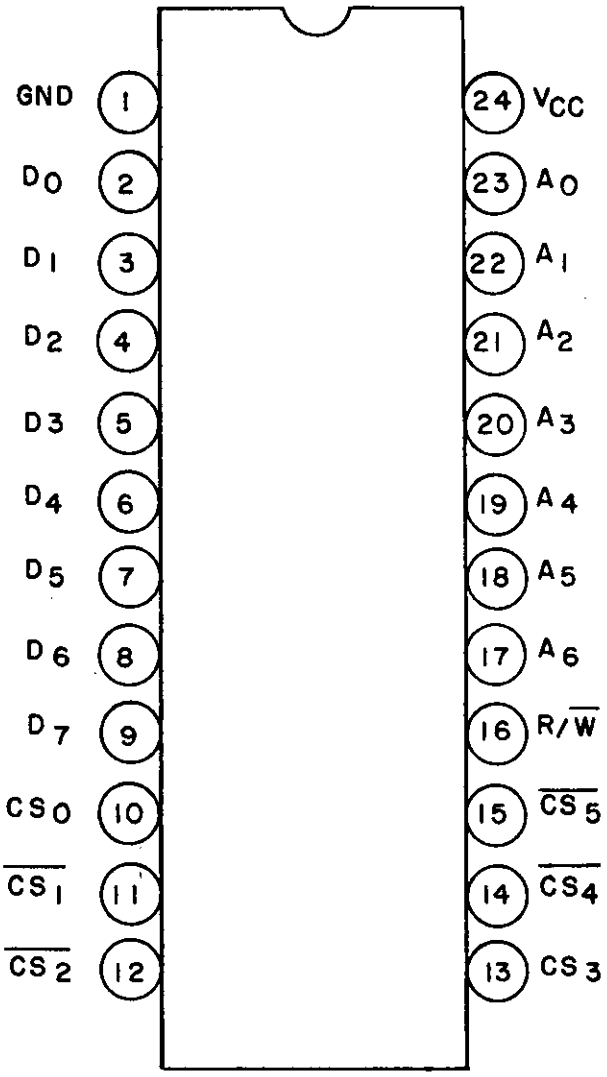


FIGURE 1. Terminal connections.

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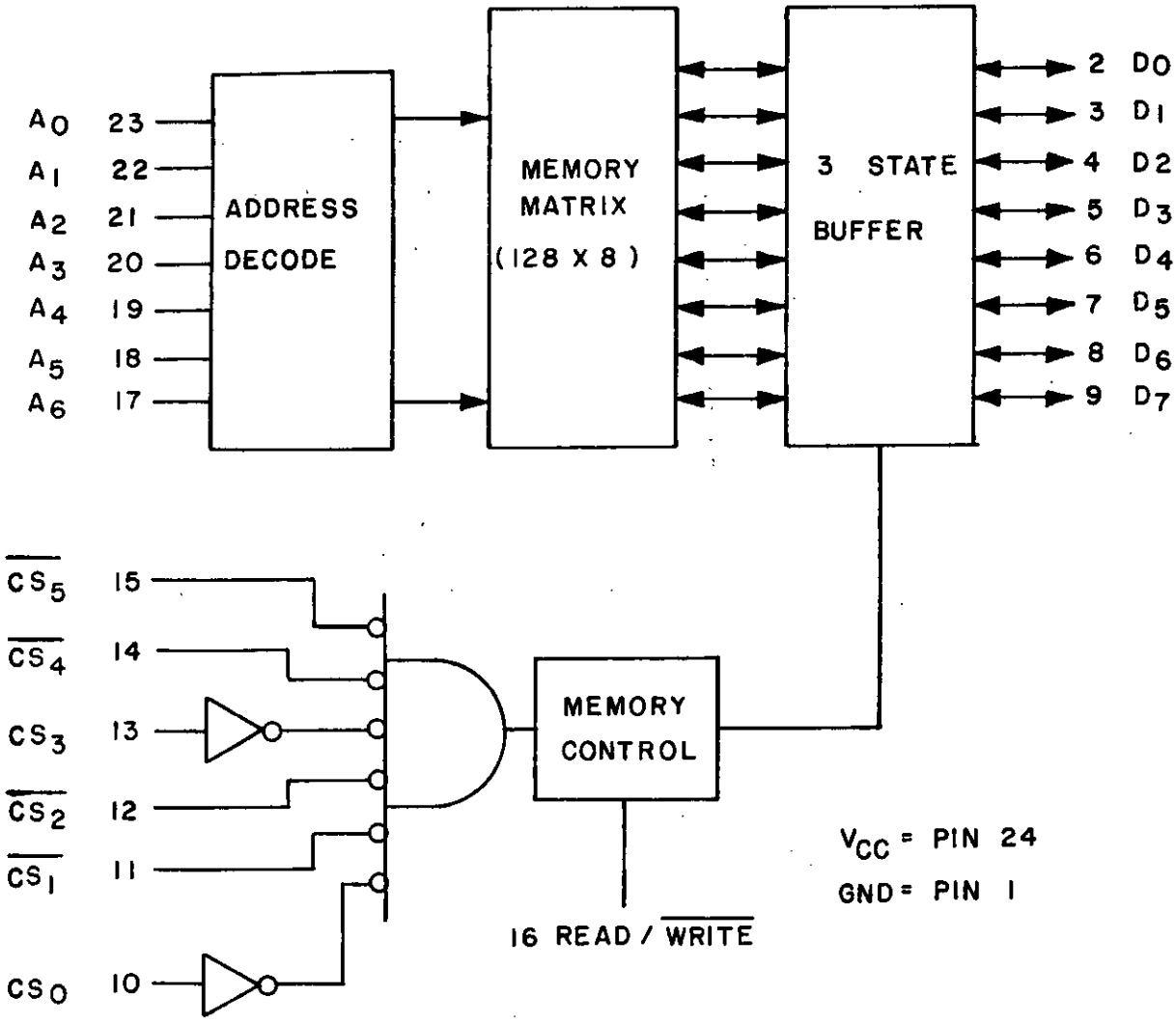


FIGURE 2. Block diagram.

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Mode 1/	R/$\overline{W}$	CS_0, CS_3	$\overline{CS}_1$, $\overline{CS}_2$, $\overline{CS}_4$, $\overline{CS}_5$,
Write	L	H	L
Read	H	H	L
Not Selected	X	L	X
Not Selected	X	X	H

1/ All chip selected inputs noted in the colum headings must be in the state indicated for proper mode selection.

Key:

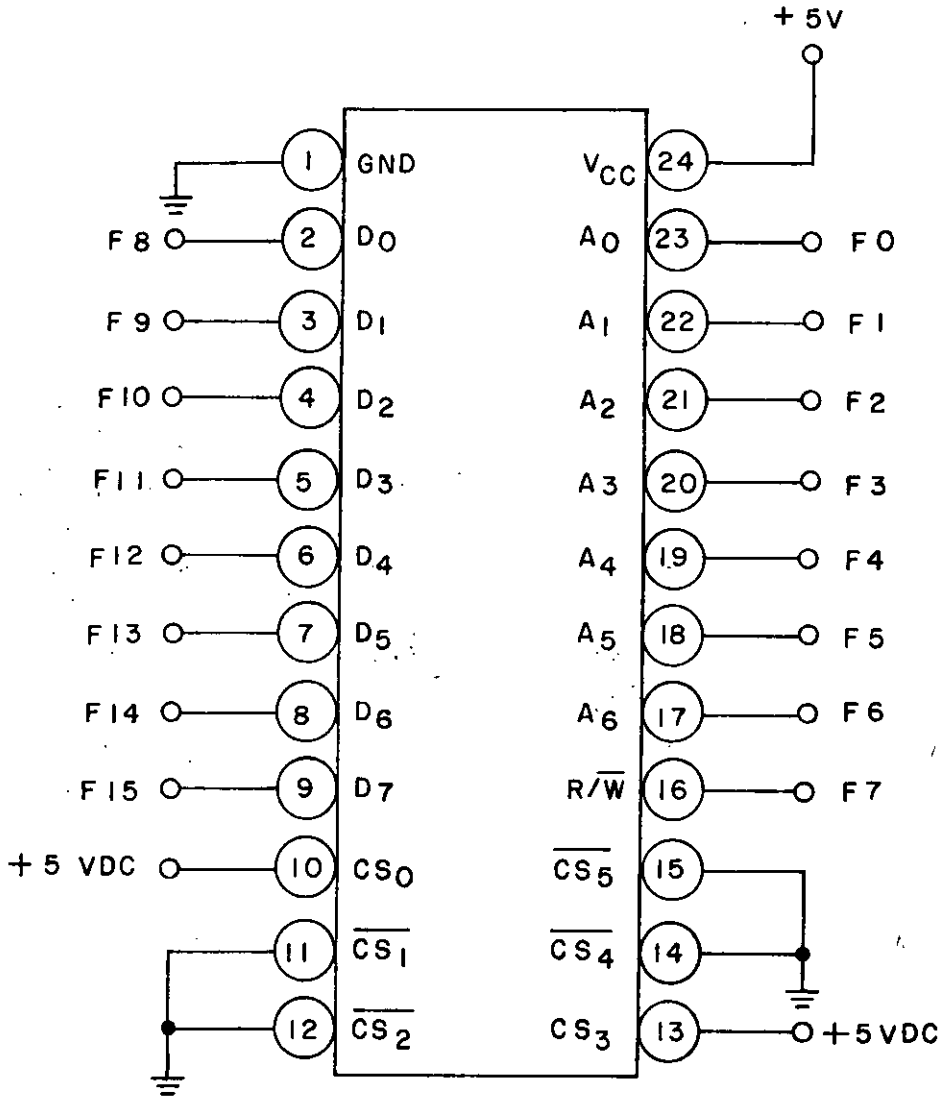
L = Low = 0 = logic false

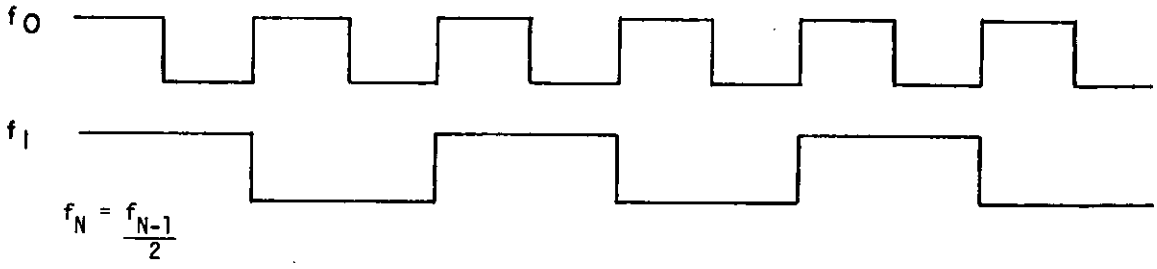
H = High = 1 = logic true

X = Don't care

FIGURE 3. Truth table.

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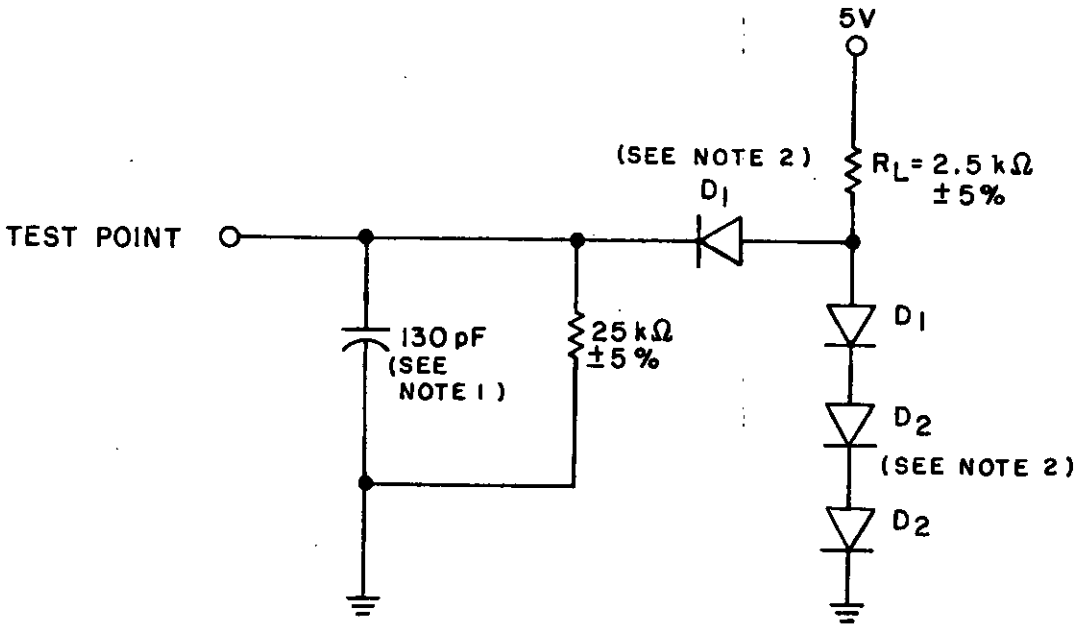




NOTES:
1. f_0 = 1 MHz, 50% duty cycle.
2. Address to be cycled in a binary sequence where A_0 is least significant and A6 is most significant.

FIGURE 4. Burn-in and steady state life test circuit and waveforms.

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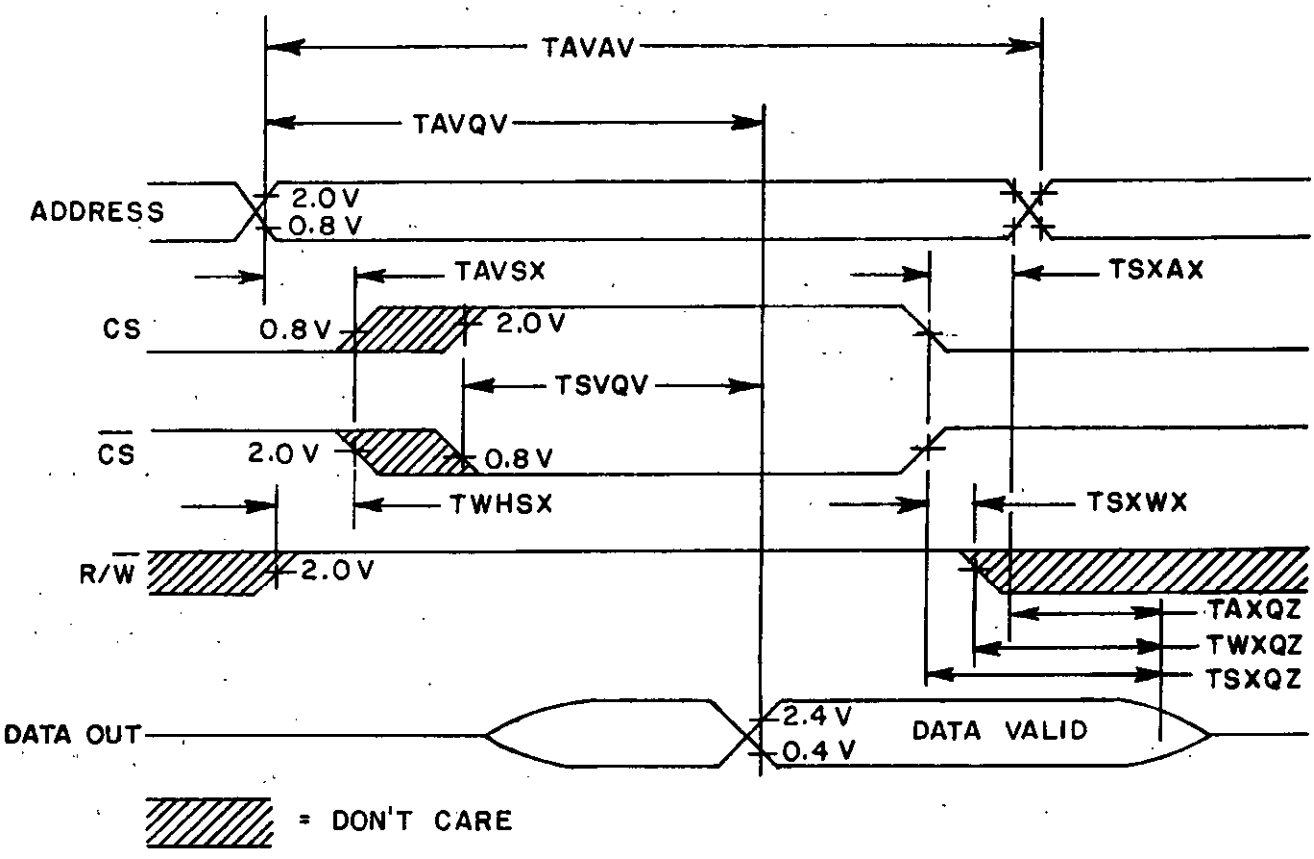


NOTES:
1. Includes Jig capacitance.
2. D1 = MMD 6150 or equivalent.
   D2 = MMD 7000 or equivalent.

FIGURE 5. AC test load.

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READ CYCLE TIMING (SEE NOTES 1 AND 2)

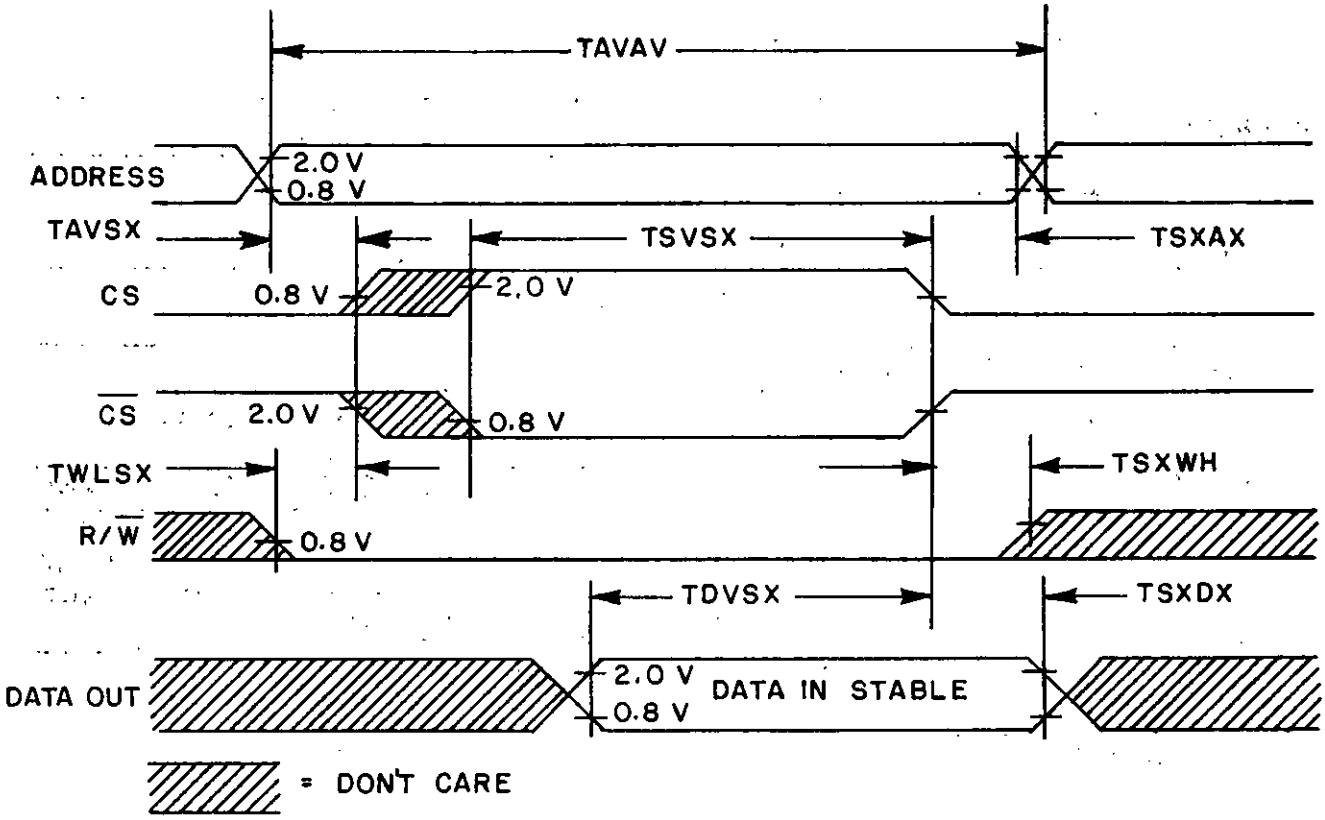


NOTES:
1. CS and $\overline{CS}$ can be enabled for consecutive read cycles provided R/$\overline{W}$ remains at V_{IH}.
2. See table I for limits.

FIGURE 6. Read cycle waveforms and test conditions.

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WRITE CYCLE TIMING (SEE NOTES 1 AND 2)



NOTES:
1. CS and $\overline{CS}$ can be enabled for consecutive write cycles provided R/$\overline{W}$ is strobed to V_{IH} before or coincident with the Address change and remains high for time TAVSX.
2. See table I for limits.

FIGURE 7. Write cycle waveforms and test conditions.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high, low, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Algorithms 1/	Measured terminal	Test limits Min	Test limits Max	Unit
				GND	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	$\overline{CS}_0$	$\overline{CS}_1$	$\overline{CS}_2$	CS_3	$\overline{CS}_4$	$\overline{CS}_5$	R/W	A_6	A_5	A_4	A_3	A_2	A_1	A_0	V_{CC}					
1 T_C = 25°C	V_{OH} 2/	3006	1 2 3 4 5 6 7 8	GND	-205 µA	-205 µA	-205 µA	-205 µA	-205 µA	-205 µA	-205 µA	-205 µA	2.0 V	0.8 V	0.8 V	2.0 V	0.8 V	0.8 V	2.0 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	4.5 V		D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7	2.4		V
	V_{OL} 2/	3007	9 10 11 12 13 14 15 16		1.6 mA	1.6 mA	1.6 mA	1.6 mA	1.6 mA	1.6 mA	1.6 mA	1.6 mA																	D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7		0.4	
	I_{IH}	3010	17 18 19 20 21 22 23 24 25 26 27 28 29 30										5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V	5.5 V		$\overline{CS}_0$ $\overline{CS}_1$ $\overline{CS}_2$ $\overline{CS}_3$ $\overline{CS}_4$ $\overline{CS}_5$ R/$\overline{W}$ A_6 A_5 A_4 A_3 A_2 A_1 A_0		2.5	µA
	I_{IL}	3009	31 32 33 34 35 36 37 38 39 40 41 42 43 44																								GND		$\overline{CS}_0$ $\overline{CS}_1$ $\overline{CS}_2$ $\overline{CS}_3$ $\overline{CS}_4$ $\overline{CS}_5$ R/$\overline{W}$ A_6 A_5 A_4 A_3 A_2 A_1 A_0			
	I_{LO1}		45 46 47 48 49 50 51 52		0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.8 V	2.0 V	2.0 V	0.8 V	2.0 V	2.0 V	0.8 V										D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7		10	
	I_{LO2}		53 54 55 56 57 58 59 60		2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V																D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7			
	I_{CC}	3005	61	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND										V_{CC}		100	mA

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high, low, or open).

Subgroup	Symbol	MIL-STD-883 method	Case V / Test no.	1 GND	2 D_0	3 D_1	4 D_2	5 D_3	6 D_4	7 D_5	8 D_6	9 D_7	10 $\overline{CS}_0$	11 $\overline{CS}_1$	12 $\overline{CS}_2$	13 CS_3	14 $\overline{CS}_4$	15 $\overline{CS}_5$	16 R/$\overline{W}$	17 A_6	18 A_5	19 A_4	20 A_3	21 A_2	22 A_1	23 A_0	24 V_{CC}	Algorithms 1/	Measured terminal	Test limits: Min	Test limits: Max	Unit
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C.																														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C.																														
7 T_C = 25°C 3/	TAVQV	Fig. 6,7	62	GND	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4.5 V	CKBD & $\overline{CKBD}$	All D_{OUT}	V_{OL} <1.5	V_{OH} >1.5	V
			63		5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/						
			64		6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/						
			65		7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/						
	TSVQV		66																												
	TAVQV		67																												
			68																								MARCH DIAG CKBD & $\overline{CKBD}$					
			69		4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/						
			70		4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	5.5 V					
			71		5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	4.5 V					
			72		5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5.5 V					
			73		6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	4.5 V					
			74		6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	5.5 V					
			75		7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	4.5 V					
			76																							5.5 V					
	TSVQV		77																							4.5 V					
	TSVQV		78																							5.5 V					
	TAVQV		79																							4.5 V					
			80																							5.5 V					
			81																							4.5 V	MARCH DIAG					
			82																							5.5 V	MARCH DIAG					
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = 125°C and -65°C.																														
9 T_C = 25°C 3/	TAVQV	Fig. 6,7	83	GND	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	4.5 V	GALPAT & GALRESH	All D_{OUT}		450	ns
	TAVQV		84																							5.5 V				450	
	TSVQV		85																							4.5 V				230	
	TSVQV		86																							5.5 V				230	
	TWLQZ		87																							4.5 V			10	80	
	TWLQZ		88																							5.5 V				80	
	TSLQZ		89																							4.5 V					
	TSLQZ		90																							5.5 V					
	TALQZ		91																							4.5 V					
	TALQZ		92																							5.5 V					

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high, low, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J: Test no.	1: GND	2: D_0	3: D_1	4: D_2	5: D_3	6: D_4	7: D_5	8: D_6	9: D_7	10: CS_0	11: $\overline{CS}_1$	12: $\overline{CS}_2$	13: CS_3	14: $\overline{CS}_4$	15: $\overline{CS}_5$	16: R/W	17: A_6	18: A_5	19: A_4	20: A_3	21: A_2	22: A_1	23: A_0	24: V_{CC}	Algorithms 1/	Measured terminal	Test limits: Min	Test limits: Max	Unit
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = +125°C.																														
11	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = -55°C.																														

1/ See appendix for description of algorithms.

2/ For V_{OH} and V_{OL} measurements, the address indicated should have been previously written for output high (V_{OH}) and then rewritten to V_{OL}, unless a different address is used where known output states can be selected.

3/ All inputs for functional testing shall be V_{IH} = 2.0 V and V_{IL} = 0.8 V.

4/ All write pulse timing shall be set to nominal when writing before doing read access time measurements.

5/ Set write pulses TAVSX, TSVSX, TSXAX, and TAVAV to minimum prior to doing a "Read" test.

6/ In addition to the minimum write conditions of 5/, set TWLSX and TSXWH to minimum prior to doing a "Read" test.

7/ In addition to the minimum write conditions of 5/ and 6/, set TDVSX and TSXDX to minimum prior to doing a "Read" test.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

a. A special subgroup shall be added using a lot tolerance percent defective of 15 for class B devices. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

b. The electrical tests shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical tests shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A inspection. These additional tests shall consist of performing the specified tests using the GALPAT pattern and then repeating using the GALRESH pattern as specified in the appendix.

c. Steady-state life test (method 1005 of MIL-STD-883, test condition D or F) using the circuit shown on figure 4, or equivalent.

(1) T_A = 125°C minimum.

(2) Test duration: 1,000 hours except as permitted by appendix B of MIL-M-38510.

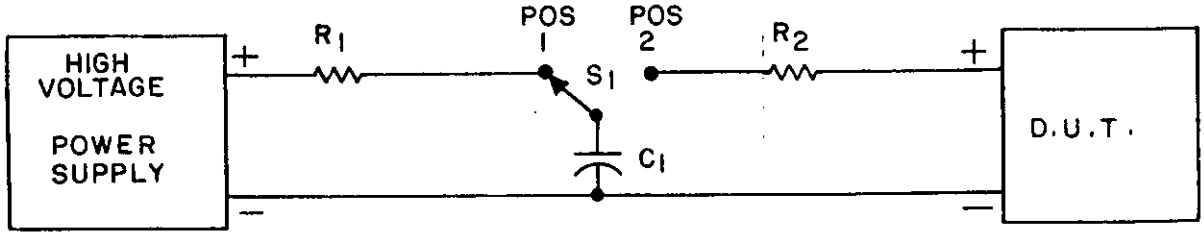
4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the steady state life or burn-in test condition, they shall be cooled to 35°C or less, prior to removal of bias.

4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of four) of the device under test (DUT) shall be subjected to a voltage pulse from a 100-pF source charged to 170 volts. This destructive test shall be conducted in accordance with the following, using the test circuit shown on figure 8.



V_{ZAP} = 170 V minimum charge on C1.
10 MΩ ≤ R1 ≤ 50 MΩ.
R2 = 1.5 kΩ
C1 = 100 pF ±20%
S1 = Hg - wetted "bounceless" relay.

FIGURE 8. High voltage (V_{ZAP}) test circuit.

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a. Measure I_{IL} and I_{IH} at inputs selected as stated above, at 25°C. The input conditions shall be in accordance with table III. The test limit for each input tested shall be ±10 microamperes maximum.

b. In the circuit shown, charge the capacitor to +170 volts. Then, using the same terminal of the device as selected above for leakage measurements, switch the capacitor to discharge into the device terminal. Repeat the procedure with -170 volts.

c. Within 24 hours, repeat the I_{IL} and I_{IH} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be as specified in MIL-M-38510. The devices covered by this specification require electrostatic protection.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity if applicable.

e. Requirements for packaging and packing.

f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

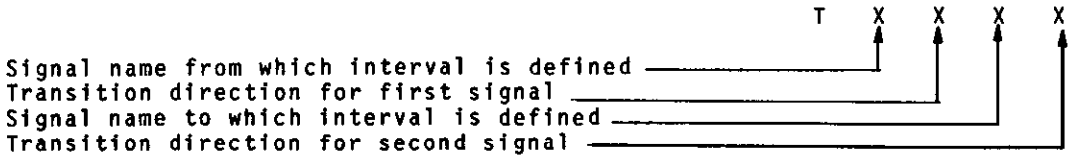
6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

A_0 thru A_6	Address inputs.
CS	Chip select, input.
D_0 thru D_7	Data inputs/data outputs.
GND	Electrical ground (common terminal).
$R/\overline{W}$	Read or write select input.
TAVAV	Read or write cycle time.
TAVQV	Address access time.
TAVSX	Address setup time prior to CS.
TAXQZ	Data hold time from end of address change.

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TDVSX	Data setup time.
TSVQV	Chip select access time.
TSVSX	Chip select pulse width.
TSXAX	Address hold time from end of CS.
TSXDX	Data hold time from end of CS.
TSXQZ	Data valid from end of CS.
TSXWH	Write hold time from end of CS.
TSXWX	Read hold time from end of CS.
TWHSX	Read to chip select time.
TWLSX	Write to CS time.
TWXQZ	Data hold time from end of read.

6.4.1 Timing parameter abbreviations. All timing abbreviations use uppercase characters with no subscripts. The initial character is always "T" and is followed by four descriptors. These characters specify two signal points arranged in a "from-to" sequence that define a timing interval. The two descriptors for each signal point specify the signal name and the signal transitions. Thus the format is:



a. Signal definitions:
   - A = Address
   - D = Data in
   - E = Chip enable
   - G = Output enable
   - Q = Data out
   - S = Chip select
   - W = Write enable

b. Transition definitions:
   - H = Transition to high
   - L = Transition to low
   - V = Transition to valid
   - X = Transition to invalid or don't care
   - Z = Transition to off (high impedance)

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.6 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D generic data shall be on date codes no more than one year old and on the same package type (see 3.1.3.12 of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

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6.7 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	6810

6.9 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.10 Changes from previous issue. Asterisks are not used in this revision to indentify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0603-2)

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APPENDIX

FUNCTIONAL ALGORITHMS

Functional algorithms are test patterns which define the exact sequence of tests used to verify proper operation of a random access memory microcircuit. Each algorithm serves a specific purpose for testing the device.

10. FUNCTIONAL PATTERNS.

10.1 Pattern 1 - CKBD.

a. Write a checkerboard pattern into memory ("0" in address "0") from address "0" to "N".

b. If the CS off test is to be performed, attempt to write the complement pattern into the cell memory with the device not selected.

c. Read checkerboard pattern in the memory.

10.2 Pattern 2 - $\overline{\text{CKBD}}$. Same as CKBD only with data complemented.

10.3 Pattern 3 - MARCH.

a. Write test word into every location.

b. The addressing is then scanned from location "0" to location "N".

c. At each address, the test word is read and a complemented test word is written back into the same location.

d. The addressing is then scanned in reverse from location "N" to location "0".

e. At each address, the complemented test word is read and the test word is written back in.

10.4 Pattern 4 - SHIFTING DIAGONAL.

a. Write a background of "0's" except for one diagonal line of "1's".

b. Read entire memory, then reload memory with the diagonal stripe of "1's" shifted one cell to the right.

c. Repeat b until the diagonal stripe has reappeared on the other side of the memory and eventually shift to the original start position.

d. Repeat a through c but complement the pattern. A diagonal line of "0's" in a background of "1's".

10.5 Pattern 5 - GALPAT. This program will test all bits in the array, the addressing, and interaction between bits for ac performance. The memory is initialized by writing a field of "1" and then a field of "0" into the cell memory.

Description:

a. Write a "1" in word "0" (reference location).

b. Word "0" is read.

c. Word "1" is read.

d. Word "0" is read.

e. Word "2" is read.

f. Word "0" is read.

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g. The reading procedure continues back and forth between word "0" and the next higher numbered word until word "127" is reached, then increments to the next word (which becomes the reference location). Steps a through g are then repeated until all the words in the memory are used at least once as the reference location.

10.6 Pattern 6 - DIAGONAL GALRESH (with row column ping pong read GGII). This pattern will test all bits in the array for writing instruction and switching performance.

a. Initialize the memory by writing a field of "0's".

b. Perform the following read/write sequence moving the test word along the diagonal of the memory and reading only the row and column of the test word in ping pong fashion:

R0 = Read "0" W1 = Write "1" etc.

	Step								
	1	2	3	4	5	6	7	8	9
BACKGROUND WORD		R0		R0		R0		R0	
TEST WORD	R0		W1		R1		W0		R0

c. Reinitialize the memory by writing a field of "1's".

d. Perform the following read/write sequence moving the test bit along the diagonal of the memory and reading only the row and column of the test bit in ping pong fashion:

	Step								
	1	2	3	4	5	6	7	8	9
BACKGROUND BIT		R1		R1		R1		R1	
TEST BIT	R1		W0		R0		W1		R1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/402A	MICROCIRCUITS, DIGITAL, MOS, 1024-BIT STATIC RANDOM ACCESS MEMORY (RAM), MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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